

Name: Farid Javed Shaikh

Roll no: B-36

Branch:- AI&DS(B)

Experiment:-06

AIM:- Implementing Web Scraping in Python with BeautifulSoup

Theory :-

1.What is web scraping? Role of Requests library Web scraping is the process of extracting data from websites automatically using programs.

Role of Requests library:

Sends HTTP requests to web pages.

Retrieves the HTML content of a webpage.

Acts as the first step in scraping (getting the page data)

Example:-

```
import requests
url = "https://example.com"
response = requests.get(url)
print(response.text)
```

```
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="vi
```

2.What is BeautifulSoup? How does it help?

BeautifulSoup is a Python library used to parse HTML and XML documents.

How it helps:

Converts raw HTML into a structured format (parse tree).

Makes it easy to search and extract elements

Handles poorly formatted HTML.

Example:-

```
from bs4 import BeautifulSoup
html = "<html><body><h1>Hello</h1></body></html>"
```

```
soup = BeautifulSoup(html, "html.parser")
print(soup.h1.text)
```

Hello

3. Difference between Web Scraping and Web Crawling

ANS: Web scraping and web crawling are related but serve different purposes in handling web data. Here's a clear comparison:

1. Definition Web Crawling: The process of automatically browsing the internet to discover and index web pages. It systematically follows links from one page to another. Web Scraping: The process of extracting specific data from web pages (like text, prices, images, etc.).
2. Purpose Web Crawling: Used mainly by search engines like Google or Bing to index websites. Web Scraping: Used to collect targeted data for analysis, price comparison, research, etc.
3. Scope Web Crawling: Broad—focuses on discovering many pages across the web. Web Scraping: Narrow—focuses on extracting specific information from selected pages.
4. Process Web Crawling: Starts with a list of URLs (seed URLs), visits them, and follows links recursively. Web Scraping: Accesses specific pages and parses their content to extract required data.
5. Output Web Crawling: Produces an index or list of URLs and metadata. Web Scraping: Produces structured data (like CSV, JSON, or database entries).
6. Example Tools Web Crawling: Apache Nutch, Heritrix Web Scraping: BeautifulSoup, Scrapy
7. Relationship

Web crawling often comes before web scraping. A crawler finds the pages, and a scraper extracts the data from them.

4. Commonly used methods in BeautifulSoup

Some important methods:

`find()` → Finds first matching element.

`find_all()` → Finds all matching elements.

`get_text()` → Extracts

`text.get()` → Gets attribute value

`select()` → CSS selector-based search

Example:-

```
soup.find('p')  
soup.find_all('a')  
soup.get_text()
```

```
'Hello'
```

5.Ethical and Legal Considerations

When doing web scraping, you must follow rules:

Ethical considerations:

Respect website policies (check robots.txt).

Avoid overloading servers (use delays).

Do not scrape sensitive/private data

Legal considerations:

Follow terms of service of websites.

Avoid copyrighted or restricted content.

Ensure data is used responsibly

List item

```
import requests  
from bs4 import BeautifulSoup  
  
# Step 1: Define the URL  
url = "https://example.com"  
  
# Step 2: Send HTTP request  
response = requests.get(url)  
  
# Step 3: Check if request was successful  
if response.status_code == 200:  
    print("Successfully fetched the webpage!\n")  
  
    # Step 4: Parse HTML content  
    soup = BeautifulSoup(response.text, "html.parser")  
  
    # Step 5: Extract all links (anchor tags)  
    links = soup.find_all('a')  
  
    print("Links found on the webpage:\n")  
  
    for link in links:
```

```
for link in links:
    text = link.get_text()
    href = link.get('href')
    print("Text:", text)
    print("Link:", href)
    print("-" * 40)
else:
    print("Failed to retrieve webpage. Status code:", response.status_code)
```

Successfully fetched the webpage!

Links found on the webpage:

Text: Learn more

Link: <https://iana.org/domains/example>

Conclusion:-

Web scraping using tools like Requests and BeautifulSoup enables efficient data extraction, but must always be done responsibly, ethically, and legally. This practical demonstrated how to programmatically fetch web content and parse it to extract specific information, such as links. Web scraping is a powerful technique for data collection from the internet, vital for tasks like market research, data analysis, and content aggregation, provided it adheres to legal frameworks and website policies.